

```
1  #include<stdio.h>
2  int
3  main()
4  {
5  int bt[20],p[20],wt[20],tat[20],i,j,n,total=0,pos,temp;
6  float avg_wt,avg_tat;
7  printf("Enter number of process:");
8  scanf("%d",&n);
9  printf("\nEnter Burst Time:\n");
10 for(i=0;i<n;i++)
11 {
12 printf("p%d:",i+1);
13 scanf("%d",&bt[i]);
14 p[i]=i+1;
15 }
16 for(i=0;i<n;i++){
17 pos=i;
18 for(j=i+1;j<n;j++)
19 {
20 if(bt[j]<bt[pos])
21 pos=j;
22 }
23 temp=bt[i];
24 bt[i]=bt[pos];
25 bt[pos]=temp;
26
27 temp=p[i];
28 p[i]=p[pos];
29 p[pos]=temp;
30 }
31 wt[0]=0;
32 for(i=1;i<n;i++)
33 {
34 wt[i]=0;
35 for(j=0;j<i;j++)
36 wt[i]+=bt[j];
37
38 total+=wt[i];
39 }
40 avg_wt=(float)total/n;
41 total=0;
```

Output

Code Execution Successful

Sample input is shown on the next page.

SJF CPU Scheduling – Sample Run

Input:

Enter number of process: 4

Enter Burst Time:

p1: 6

p2: 8

p3: 7

p4: 3

Processes are arranged by
Shortest Job First (SJF).

Output

Enter number of process:4

Enter Burst Time:

p1:6

p2:8

p3:7

p4:3

Process	Burst Time	Waiting Time	Turnarou
p4	3	0	3
p1	6	3	9
p3	7	9	16
p2	8	16	24

Average Waiting Time=7.000000

Average Turnaround Time=13.000000

=== Code Execution Successful ===